MILITARY SPECIFICATION SHEET

# CABLE, ELECTRICAL, 300 VOLTS, TYPE TPNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

## Construction

(B) First - Copper conductor, uncoated, Navy Standard size 3/5(7).
Second - Semi-rigid polyvinyl chloride 80° Celsius (C) insulation, 0.009-inch nominal (nom) thickness. Telephone Identification Code applied by method 3.
Third - Two conductors cabled together to form a pair. Maximum (max) lay 2.0 inches.
Fourth - The required number of pairs (see Details) cabled with a lay not greater than 10 inches. Cabling sequence shall be consecutive, starting with pair No. 1, from center outward. Fillers as necessary in order to form a firm, well-rounded assembly. Fillers shall be fibrous.
Fifth - Binder tape of 0.002-inch thick corrugated polyester type G of MIL-I-631.
Sixth - Special thermoplastic jacket (gray). See Details for thickness. Cable surface marking required.

## Details

| Type and Size | Number of pairs | Cable jacket thickness, nom (inch) | Overall diameter, max (inch) | Cold bending mandrel diameter, max (inches) |
|---|---|---|---|---|
| TPNW-1-1/2 | 1/ 1-1/2 | 0.040 | 0.22 | 3.0 |
| TPNW-3 | 3 | .040 | .29 | 3.5 |
| TPNW-5 | 5 | .040 | .34 | 4.0 |
| TPNW-10 | 10 | .040 | .45 | 5.5 |
| TPNW-15 | 15 | .040 | .51 | 6.5 |
| TPNW-20 | 20 | .040 | .56 | 7.0 |
| TPNW-30 | 30 | .050 | .68 | 8.0 |
| TPNW-40 | 40 | .050 | .75 | 9.0 |

1/ Size 1-1/2 is composed of three conductors cabled to form a triad.

(B) Denotes changes.



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6145

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EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| **Basic electrical:** | |
| Conductor resistance - Ohms/1000 ft. at 25°C, max. | 16.33 |
| Voltage withstand - Volts, root mean square (Vrms), minimum (min) | |
| Conductor to conductor. | 2000 |
| Insulation resistance - Megohms/1000 ft. | 1000 |
| **Group A:** | |
| Visual and dimensional | |
| **Group B:** | |
| Flammability - inches, max | 2 |
| Cold bending, cable - (See Details for mandrel diameter) | No damage |
| Physicals (unaged) | |
| Insulation | |
| Tensile strength - pound-force per square inch ($lbf/in^2$), min | 3500 |
| Elongation - Percent, min. | 125 |
| Durometer (Type D) hardness (ASTM-D 2240). | 70 ± 5 |
| (B) Jacket | |
| (B) Tensile strength - $lbf/in^2$, min. | 2300 |
| (B) Elongation - percent, min. | 250 |
| **Group C:** | |
| Physicals (aged) | |
| Insulation (conductor) | |
| Air oven | |
| Elongation, percent of unaged, min. | 85 |
| Jacket | |
| Hot oil immersion | |
| (B) Tensile strength - Percent of unaged, min | 75 |
| (B) Elongation - Percent of unaged, min | 50 |
| Air oven (168 hours at 136°C) | |
| (B) Tensile strength - Percent of unaged, min | 80 |
| (B) Elongation - Percent of unaged, min | 60 |
| (B) Permanence of printing (jacket) - cycles, min. | 250 |

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTHS

| Type and size | Feet (nom) |
|---|---|
| TPNW-1-1/2 through TPNW-10 | 2500 |
| TPNW-15 and TPNW-20 | 1500 |
| TPNW-30 and TPNW-40 | 1000 |

Custodians:
Army - MI
Navy - SH
Air Force - 99
Review activities:
Army - EL, AV
Navy - EC
User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0677-72)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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